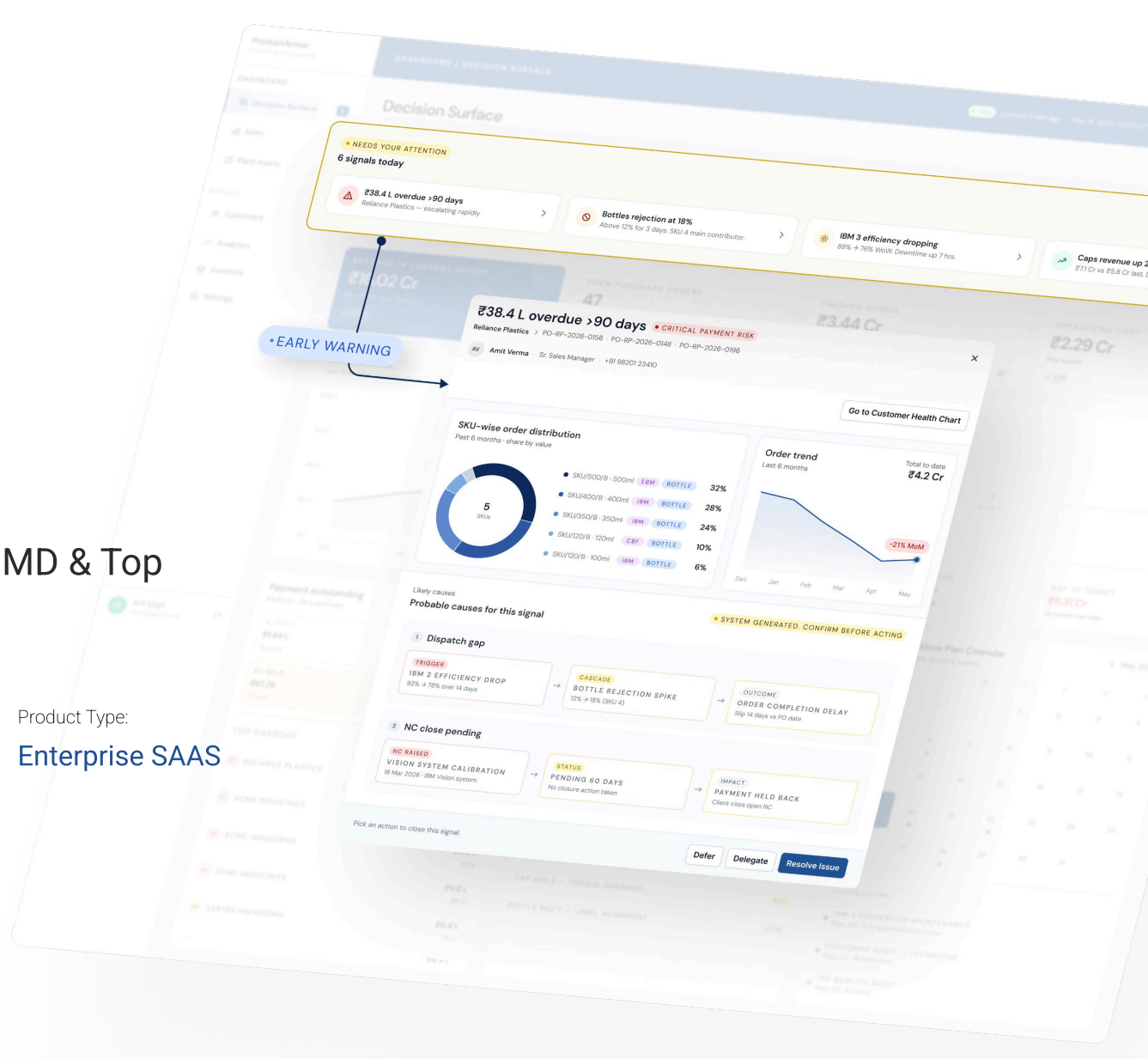
Enterprise | Atlas | Case Study 2024

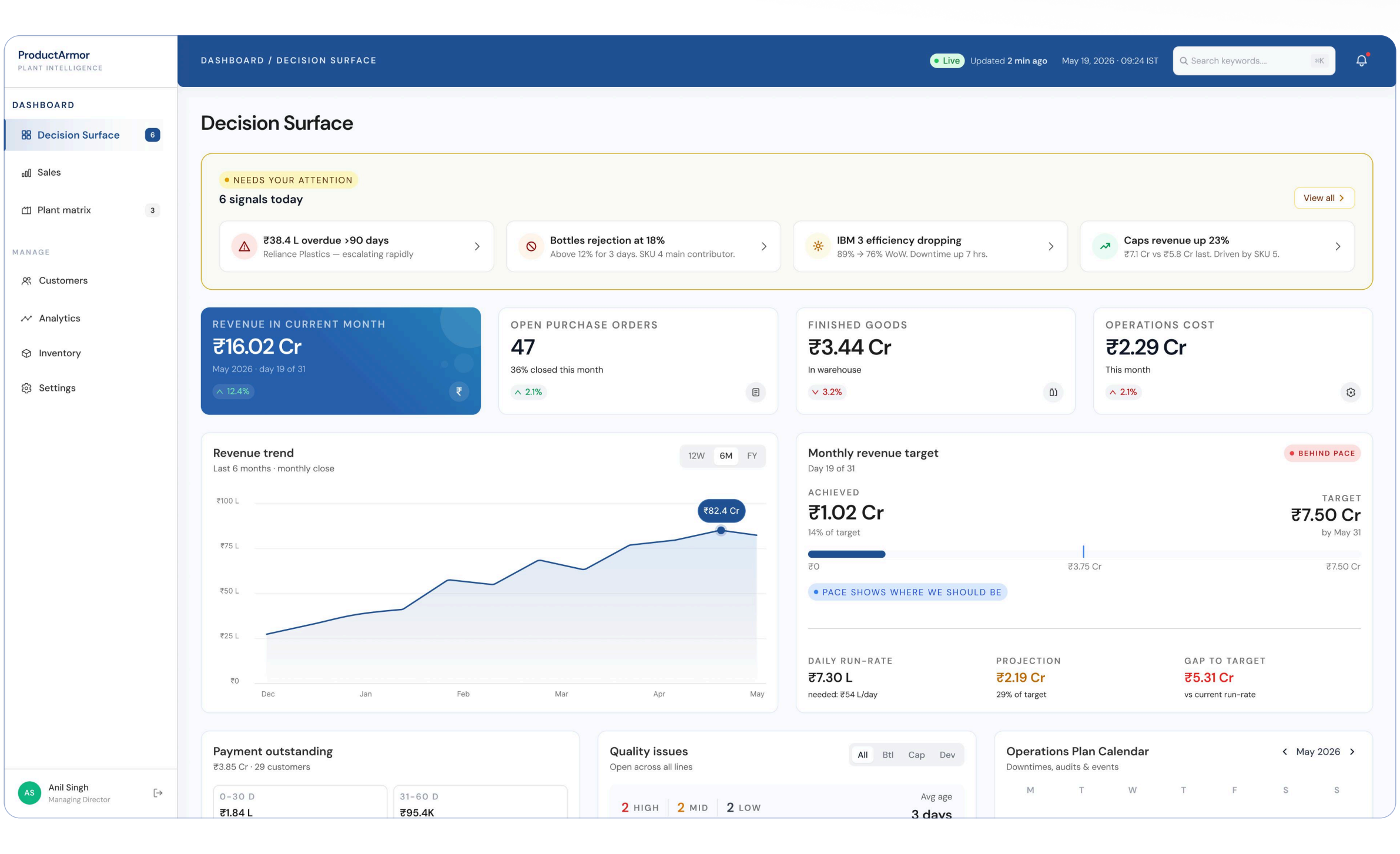
From reports to **one decision surface**


EARLY WARNING

■ Overview

Atlas is an internal **decision surface dashboard** built for the MD & Top management at Product Armor.

My Role:	Duration:	Firm:	Domain:	Product Type:
Design Lead	3 months	Product Armor	Pharmaceutical	Enterprise SAAS



ProductArmor
PLANT INTELLIGENCE

DASHBOARD / DECISION SURFACE · Live · Updated 2 min ago · May 19, 2026 · 09:24 IST

Decision Surface

NEEDS YOUR ATTENTION
6 signals today

- ₹38.4 L overdue >90 days — Reliance Plastics — escalating rapidly
- Bottles rejection at 18% — Above 12% for 3 days. SKU 4 main contributor.
- IBM 3 efficiency dropping — 89% → 76% WoW. Downtime up 7 hrs.
- Caps revenue up 23% — ₹7.1 Cr vs ₹5.8 Cr last. Driven by SKU 5.

REVENUE IN CURRENT MONTH	OPEN PURCHASE ORDERS	FINISHED GOODS	OPERATIONS COST
₹16.02 Cr	47	₹3.44 Cr	₹2.29 Cr
May 2026 · day 19 of 31	36% closed this month	In warehouse	This month
12.4%	2.1%	3.2%	2.1%

Revenue trend — Last 6 months · monthly close

Monthly revenue target — Day 19 of 31 — BEHIND PACE
ACHIEVED ₹1.02 Cr — 14% of target; TARGET ₹7.50 Cr by May 31
PACE SHOWS WHERE WE SHOULD BE
DAILY RUN-RATE ₹7.30 L — needed: ₹54 L/day; PROJECTION ₹2.19 Cr — 29% of target; GAP TO TARGET ₹5.31 Cr — vs current run-rate

Payment outstanding — ₹3.85 Cr · 29 customers
Quality issues — Open across all lines — 2 HIGH 2 MID 2 LOW — Avg age 3 days
Operations Plan Calendar — Downtimes, audits & events — May 2026

Anil Singh — Managing Director

Decision Surface
Sales
Plant matrix
Customers
Analytics
Inventory
Settings
12W
6M
FY
₹82.4 Cr
₹100 L
₹75 L
₹50 L
₹25 L
₹0
Dec
Jan
Feb
Mar
Apr
May

> “The **Signal feature** give me much needed start on what to do.”
>
> Mr. Harsha · MD, Product Armor

■ Context

Product Armor is a **GMP-regulated pharma packaging** plant in Hyderabad 100% automated production, zero human contact. This is achieved by maintaining class 5 clean room environment, online QC, automated guided vehicle and pallet loaders and State of the art manufacturing facility.

Plant at a glance

PHARMA PACKAGING · GMP-COMPLIANT · 100% AUTOMATED · 2024

01	02	03	04
100% Automation	**8 Verticals**	**53 Bottle SKUs**	**Online Quality check**
No human contact on line. Audit and QC stay human.	Drifts ~ crises, without early signals.	Bottles, Closures and Devices	Online Vision and automated QC system

> “With end to end automation and online vision system, when scaled for mass production this facility will be the first fully automated plant in India.”
>
> Mr. Harsha · MD, Product Armor · Interview, week 1

■ The Problem Space

The MD uses **hand-stitched spreadsheets** on which he **spends hours discussing and reconciling numbers leading to:**

01	02	03	04
Trust Fragmentation	**Drift Latency**	**Cognitive-load**	**Solution-space gap**
Cost ~5 hours/week in cross-functional reconciliation.	Drifts ~ crises, without early signals.	Machine level telemetry will not assist	ERPs assume humans run the factory.

> “I don't need more data. I need **one definition of every number** — with the drift surfaced before it becomes a crisis.”
>
> Mr. Harsha · MD, Product Armor · Interview, week 1

The brief was deceptively simple: replace eight fragmented vertical reports with one decision-led screen.

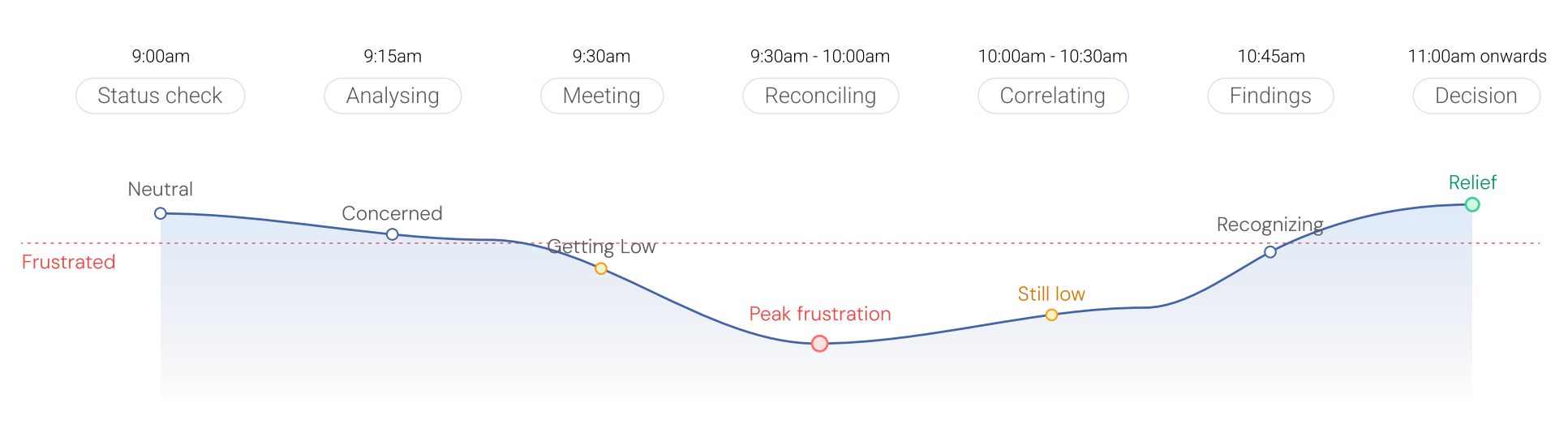
■ Approach

Week 1 - 2	Week 3	Week 4 - 8
Discover	**Strategy**	**Solution**
User Interview Journey Map Empathy Map	Goals Functional Priority	Architecture KPI Specs UI Design Interaction Delivery

■ User Research

In this case study, we used Interviews to synthesise insights gathered from MD, 4 HODs and 7 Supervisors.

User Journey Map · Empathy Map · Interviews



Pain Points

01	02	03	04	05	06
45m	**8 reports**	**0 connect**	**~25 d RCA**	**24+ NCs**	**12+ Slip**
Time to decision	Cross-vertical gap causing confusion	No cross correlation regarding issue ownership	RCA takes around 15 to 30 days	Rising NCs from client	Rising customers slippage

> “When the system is autonomous, the dashboard is the boundary surface between human judgment and machine autonomy.”
>
> Self Reflection , week 5

■ Strategy

Three principles **anchored** every subsequent decision

01	02	03
Encode Supervisory mental modal	**Design for Exceptions**	**Protect role boundaries through Architecture**
Show the Story	Show the Anomalies	Show Ownership and Accountability

■ Goals

01	02	03
Reduce decision prep-time	**Increase data-driven decision**	**Source of truth for decision making**
Attempt ~50% prep-time cut	Cross-vertical data correlation	Increase adoption for ~40% RCAs

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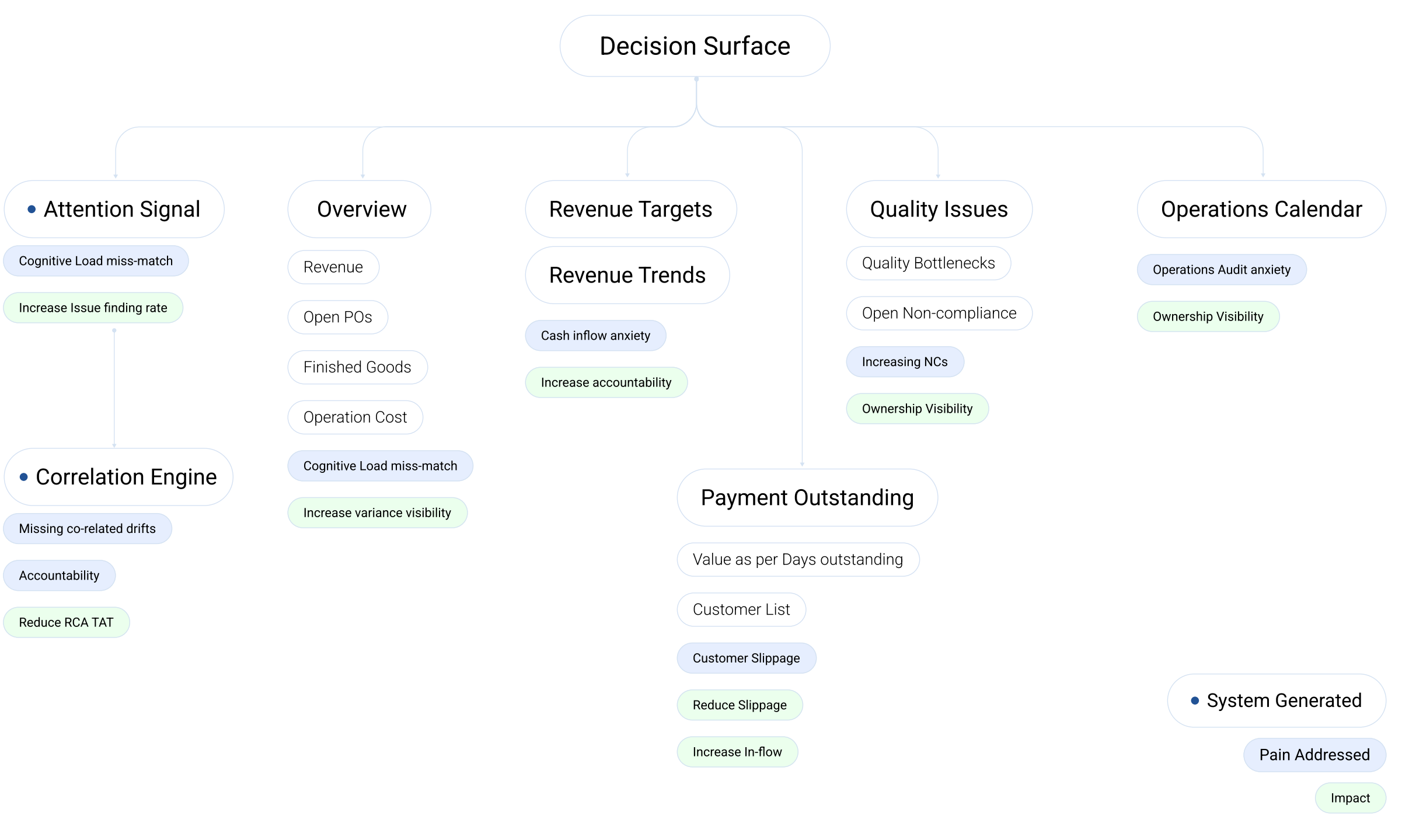
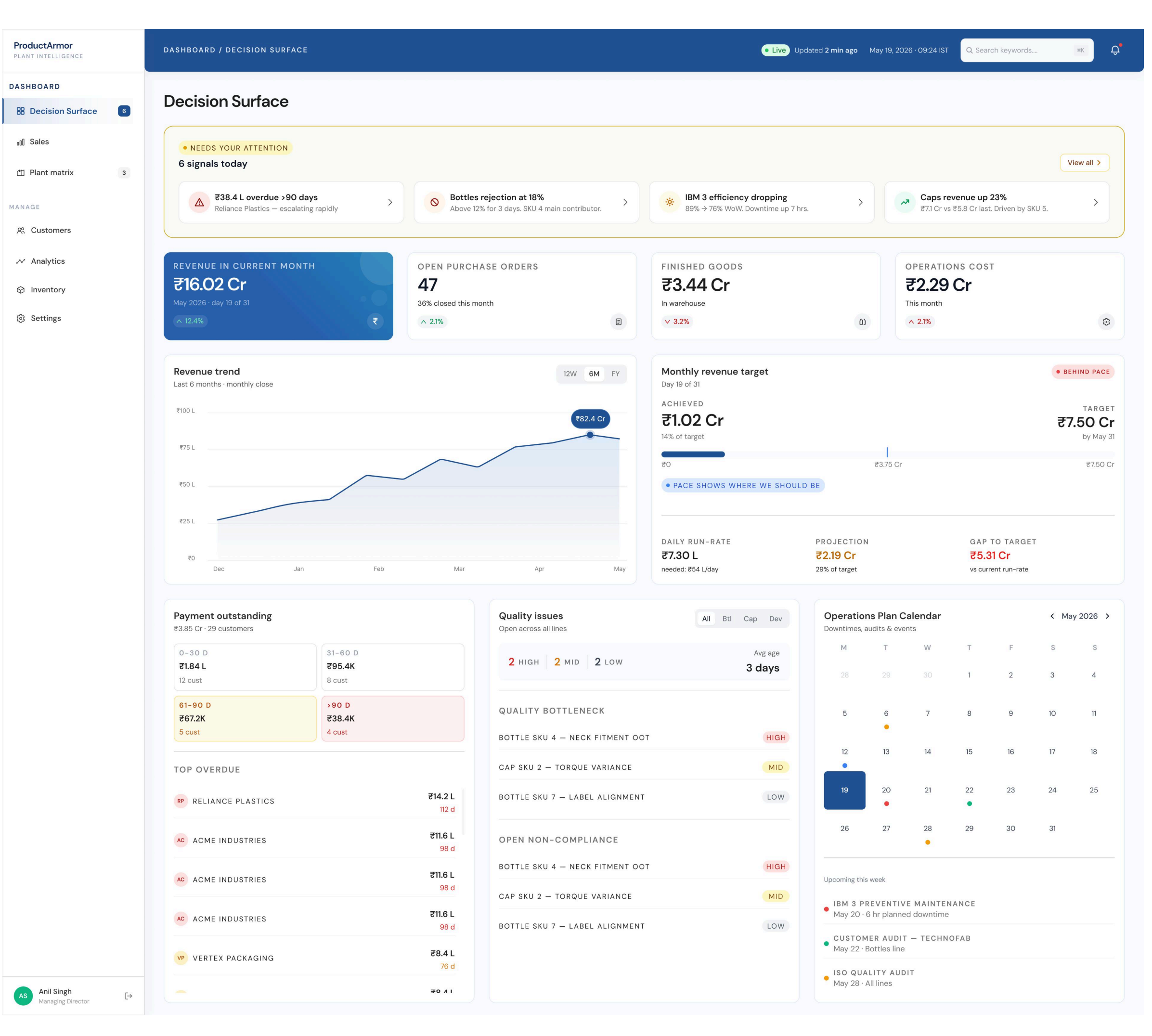
■ Hero module 1 · Decision Surface · Features

The decision hub. **Variance over Absolute**

Algorithmic attention queue Top-4 issues ranked by cross-vertical impact Show the Anomaly	**Cross-vertical correlation strip** Paired metrics drifting together surface as one connected entry. Show the Story
Active resolution interaction This makes the Attention queue active working surface. Show the Task first intent	**Variance over absolutes** Change matters more than position. Show the micro trends


Decision Surface
Attention Signal
Cognitive Load miss-match
Increase Issue finding rate
Correlation Engine
Missing co-related drifts
Accountability
Reduce RCA TAT
Overview
Revenue
Open POs
Finished Goods
Operation Cost
Cognitive Load miss-match
Increase variance visibility
Revenue Targets
Revenue Trends
Cash inflow anxiety
Increase accountability
Quality Issues
Quality Bottlenecks
Open Non-compliance
Increasing NCs
Ownership Visibility
Operations Calendar
Operations Audit anxiety
Ownership Visibility
Payment Outstanding
Value as per Days outstanding
Customer List
Customer Slippage
Reduce Slippage
Increase In-flow
System Generated
Pain Addressed
Impact



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0–30 D	31–60 D	61–90 D	>90 D
₹1.84 L	₹95.4K	₹67.2K	₹38.4K
12 cust	8 cust	5 cust	4 cust

TOP OVERDUE

RELIANCE PLASTICS	₹14.2 L	112 d
ACME INDUSTRIES	₹11.6 L	98 d
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VERTEX PACKAGING	₹8.4 L	76 d

Quality issues — Open across all lines — 2 HIGH 2 MID 2 LOW — Avg age 3 days

QUALITY BOTTLENECK
- BOTTLE SKU 4 — NECK FITMENT OOT — HIGH
- CAP SKU 2 — TORQUE VARIANCE — MID
- BOTTLE SKU 7 — LABEL ALIGNMENT — LOW

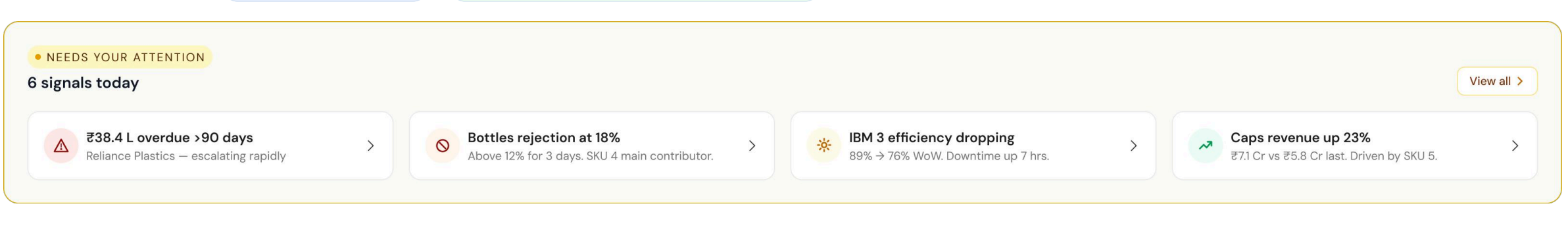
OPEN NON-COMPLIANCE
- BOTTLE SKU 4 — NECK FITMENT OOT — HIGH
- CAP SKU 2 — TORQUE VARIANCE — MID
- BOTTLE SKU 7 — LABEL ALIGNMENT — LOW

Operations Plan Calendar — Downtimes, audits & events — May 2026

Upcoming this week
- IBM 3 PREVENTIVE MAINTENANCE — May 20 · 6 hr planned downtime
- CUSTOMER AUDIT — TECHNOFAB — May 22 · Bottles line
- ISO QUALITY AUDIT — May 28 · All lines

Anil Singh — Managing Director

Attention Signal · Cognitive Load miss-match · Reduce reconciliation time from 35 mins to ~10 mins



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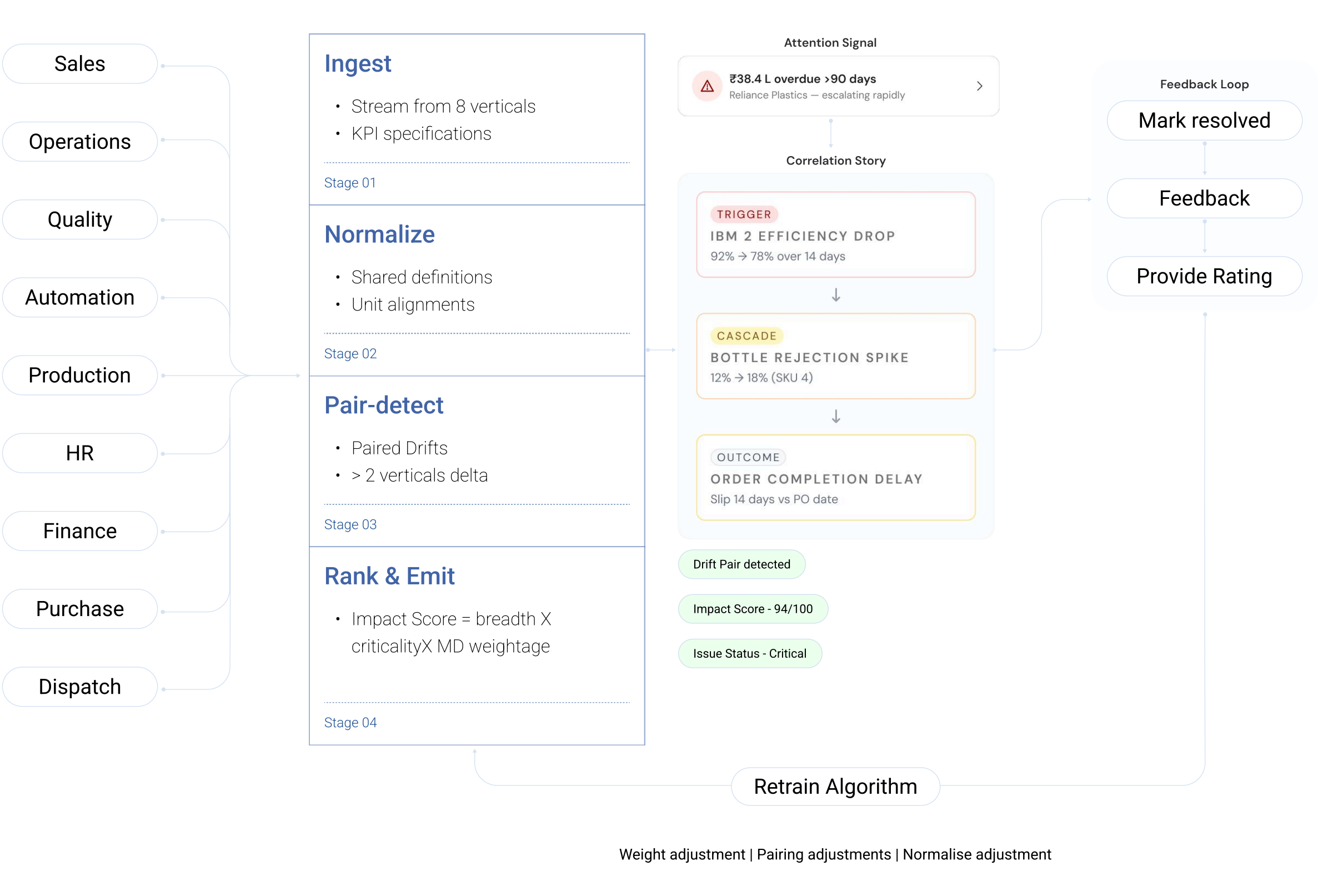
Correlation Engine Concept · Missing co-related drifts · Reduce Time -to -decision by 50%


Sales
Operations
Quality
Automation
Production
HR
Finance
Purchase
Dispatch
Ingest
Stream from 8 verticals
KPI specifications
Stage 01
Normalize
Shared definitions
Unit alignments
Stage 02
Pair-detect
Paired Drifts
> 2 verticals delta
Stage 03
Rank & Emit
Impact Score = breadth X criticalityX MD weightage
Stage 04
Attention Signal
₹38.4 L overdue >90 days
Reliance Plastics — escalating rapidly
Correlation Story
TRIGGER
IBM 2 EFFICIENCY DROP
92% → 78% over 14 days
CASCADE
BOTTLE REJECTION SPIKE
12% → 18% (SKU 4)
OUTCOME
ORDER COMPLETION DELAY
Slip 14 days vs PO date
Drift Pair detected
Impact Score - 94/100
Issue Status - Critical
Feedback Loop
Mark resolved
Feedback
Provide Rating
Retrain Algorithm

Weight adjustment | Pairing adjustments | Normalise adjustment

01	02	03
The Problem	**The Solution**	**The Trade-off**
• High Cognitive load while reconciling the number • High time to decision (~3hrs) till first decision is taken • Missing correlation KPI drift early in the process • Cross-vertical data gap	• Attention Signal providing immediate alerts to the MD • Cause engine using System intelligence for right direction of action • Providing variance to most KPIs for true company health visibility	• Instead of just numbers the new KPI setup always shows variance • Reassurance is replaced by single source of truth • Making the Overview an interactive space with attention signal at all times.

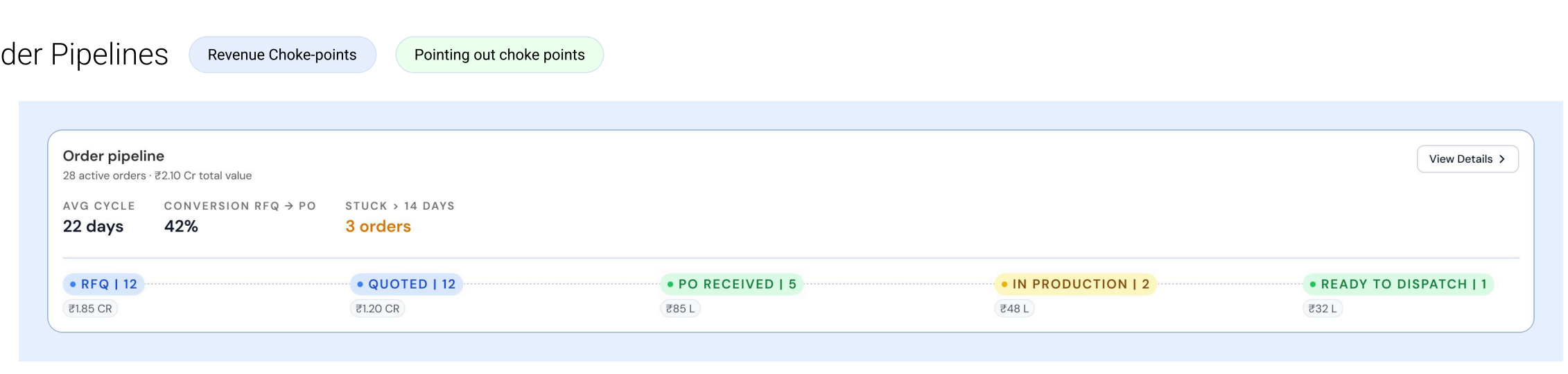
> “The **self correction mechanism** for the Cause engine is a unique feature, that will refine the system over time”
>
> Dr. Mrinmoy · HOD Automation, Product Armor

■ Hero module 2 · Sales · Features

Revenue momentum + **customer health**

Customer Health Score Top 50 customers ranked by composite score Show the Risks	**AR aging by customer** Matrix shows the customer slippage Show the Story
PO progress pipeline Provide open po matrix as a process status Show the Revenue Risk	**Sales performance matrix** Signal individual performance and revenue goals status Show the Shortfalls

Order Pipelines · Revenue Choke-points · Pointing out choke points



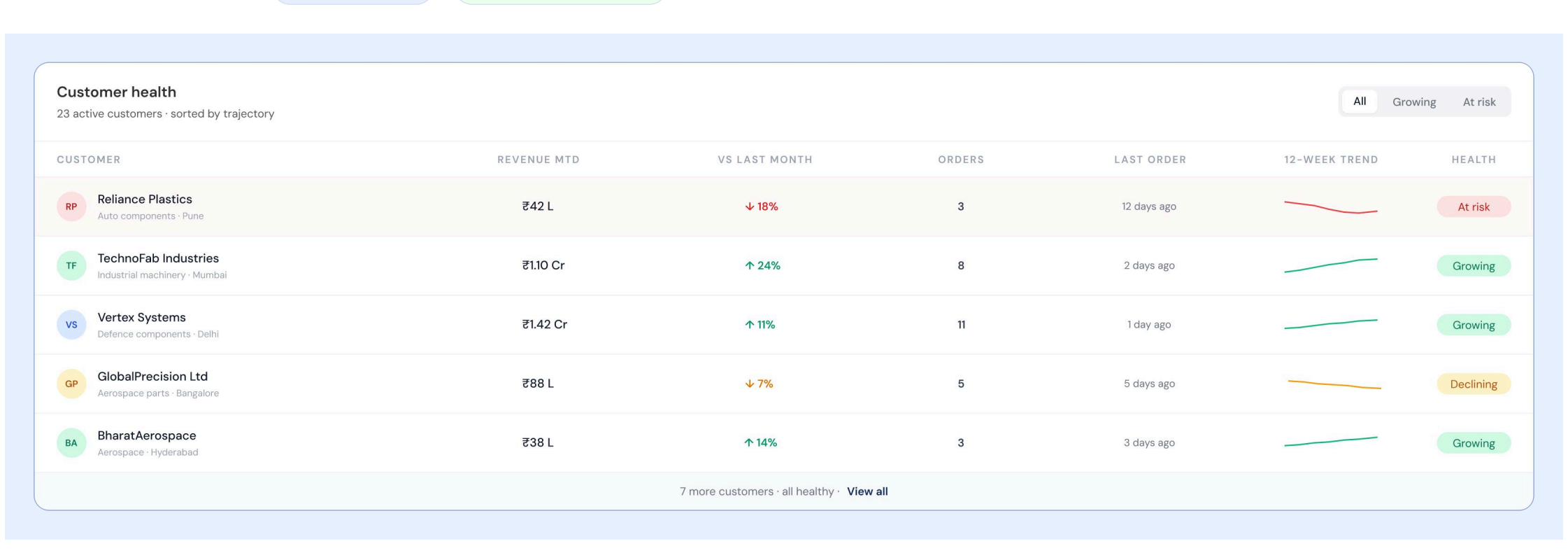
Order pipeline
28 active orders · ₹2.10 Cr total value

AVG CYCLE	CONVERSION RFQ → PO	STUCK > 14 DAYS
22 days	42%	3 orders

RFQ | 12	QUOTED | 12	PO RECEIVED | 5	IN PRODUCTION | 2	READY TO DISPATCH | 1
₹1.85 CR	₹1.20 CR	₹85 L	₹48 L	₹32 L

View Details

Customer Health Card · Cash Flow Anxiety · Focus on Risk beforehand



Customer health
23 active customers · sorted by trajectory

CUSTOMER	REVENUE MTD	VS LAST MONTH	ORDERS	LAST ORDER	12-WEEK TREND	HEALTH
Reliance Plastics Auto components · Pune	₹42 L	↓ 18%	3	12 days ago		At risk
TechnoFab Industries Industrial machinery · Mumbai	₹1.10 Cr	↑ 24%	8	2 days ago		Growing
Vertex Systems Defence components · Delhi	₹1.42 Cr	↑ 11%	11	1 day ago		Growing
GlobalPrecision Ltd Aerospace parts · Bangalore	₹88 L	↓ 7%	5	5 days ago		Declining
BharatAerospace Aerospace · Hyderabad	₹38 L	↑ 14%	3	3 days ago		Growing

7 more customers · all healthy · View all

All
Growing
At risk

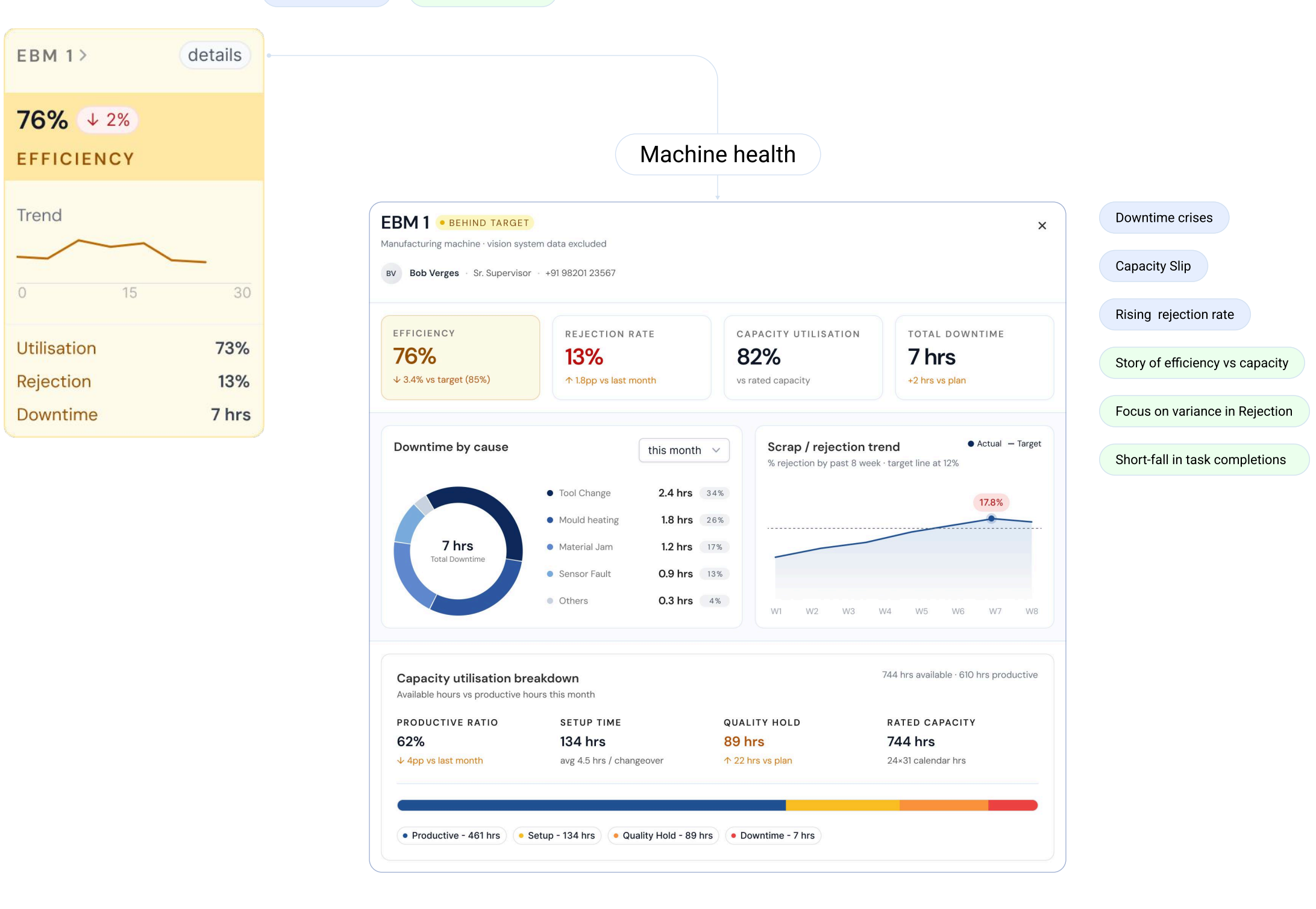
01	02	03
The Problem	**The Solution**	**The Trade-off**
• Missing Customer level focus • Cash-flow anxiety • Order Status visibility missing • Sales figure clarity	• Customer health card focuses on possible risks beforehand • PO pipeline points at the revenue bottleneck in regular PO process • Providing variance to most KPIs for true company health visibility	• Revenue MTD and AR aging should be in Decision Surface and the user will drill down to Sales module on customer level insights • Move focus from Revenue to Customer-level Insights for this module

■ Hero module 3 · Operations · Features

Where the MD **intervenes**

OEE by machine Drill down to every machine to check its health and efficiency From Capacity to Efficiency	**Rejection Variance Visibility** Showing variance across all KPIs showing exceptions. Focus on shortfall over absolute numbers
Raw material — actual vs plan Predict stocks and usage for eventualities Surface over consumption	**Audit + Maintenance** Audit and maintenance plan must go hand in hand for reduced NCs Show the micro trends

Machine Health Card · Downtime crises · Focus on short-falls



EBM 1 · details
76% ↓ 2% EFFICIENCY
Utilisation 73%
Rejection 13%
Downtime 7 hrs

Machine health

EBM 1 · BEHIND TARGET
Manufacturing machine · vision system data excluded
Bob Verges · Sr. Supervisor · +91 98201 23567

EFFICIENCY	REJECTION RATE	CAPACITY UTILISATION	TOTAL DOWNTIME
76%	13%	82%	7 hrs
↓ 3.4% vs target (85%)	↑ 1.8pp vs last month	vs rated capacity	+2 hrs vs plan

Downtime by cause — this month

Tool Change	2.4 hrs	34%
Mould heating	1.8 hrs	26%
Material Jam	1.2 hrs	17%
Sensor Fault	0.9 hrs	13%
Others	0.3 hrs	4%

Scrap / rejection trend — % rejection by past 8 week · target line at 12%

Capacity utilisation breakdown — Available hours vs productive hours this month — 744 hrs available · 610 hrs productive

PRODUCTIVE RATIO	SETUP TIME	QUALITY HOLD	RATED CAPACITY
62%	134 hrs	89 hrs	744 hrs
↓ 4pp vs last month	avg 4.5 hrs / changeover	↑ 22 hrs vs plan	24×31 calendar hrs

Productive - 461 hrs · Setup - 134 hrs · Quality Hold - 89 hrs · Downtime - 7 hrs

Trend
0
15
30
7 hrs
Total Downtime
Actual
Target
17.8%
W1
W2
W3
W4
W5
W6
W7
W8
Downtime crises
Capacity Slip
Rising rejection rate
Story of efficiency vs capacity
Focus on variance in Rejection
Short-fall in task completions

01	02	03
The Problem	**The Solution**	**The Trade-off**
• Downtime crises • Task completion visibility • Preventive maintenance accountability	• We switched the story from machine capacity utilisation to efficiency and rejection variance • Providing trend visibility for risk prediction	• Keeping Automation and Production as a separate module brings focus back to Product shortfalls and reduces confusion with automation shortfalls.

■ What we changed

The decision hub. **Variance over Absolute**

Reporting wall

- Eight verticals each producing yesterday's numbers in their own format.
- The MD spent 45 minutes every morning on reconciliation calls
- Cross-vertical drifts surfaced as separate alarms

Before · Legacy State

A decision surface

- An attention queue that ranks by cross-vertical impact.
- 10 mins to first confident decision
- Providing a holistic picture of cross-vertical drifts
- Surface slippage, variances and short-fall as probable scenarios for MD to consider

After · Atlas

■ Impact

Design to **compress decision time**

Atlas reframes the MD's morning ritual — from reconciliation calls to a glanceable decision scan.

01	02	03
~75% reduction of TTD	**80% of decision**	**Become go-to platform**
Time to decision drop from ~60 minutes of morning reconciliation calls to ~10 minutes of dashboard scan + targeted drill-down.	Lift the share of weekly decisions by ≥80% action backed by Atlas data.	Anchored to the MD's existing 7:30 AM status check ritual — Atlas becomes the first app he opens, ahead of email.
Plant Intelligence	Correlation Engine	Drift Surfacing

■ Reflections

What I'd do **differently**

Tactical

- observational study with vertical heads earlier would have surfaced the cross-functional correlation opportunities sooner.
- The Supervisors turned out to be richest source during Correlation Logic setup & pattern finding
- Understanding domain early in the design process helps.

Observational Improvement

Strategy

- The case-studies impact rests on ethnographic foundation that turned into IA decisions. I must spend more time on ethnographic research.
- Design Strategy must be free of biases and influence of the direct client for unbiased outcome.

Drawing boundaries wherever necessary

> “Industry 4.0, regulated automation, supervisory AI — the same problem shape keeps recurring. Atlas is one answer to that shape.”
>
> Why this matters!